20 In a village,

if it rains on one day, the probability that it will rain on the next day is 0.8
if it does **not** rain on one day, the probability that it will rain on the next day is 0.6

A weather forecaster says,

"There is a 70% chance that it will rain in the village on Monday."

Work out an estimate for the probability that it will rain in the village on Wednesday.
You must show all your working.

Tuesday rain = (0.7)(0.8) + (0.3)(0.6)

= 0.56 + 0.18 = 0.74.

Tuesday dry = 0.26.

Wednesday rain = (0.74)(0.8) + (0.26)(0.6)

= 0.592 + 0.156 = 0.748.

1 mark
correct two-day product

1 mark
sum all rain paths

1 mark
correct Wednesday path

1 mark
0.748

0.748

(Total for Question 20 is 4 marks)

DO NOT WRITE IN THIS AREA
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